

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

1806

Paper Title

VGG19 Transfer Learning for Leukemia Classification

Abstract

Leukemia is a hematological blood cancer disease which needs fast and accurate diagnosis to enhance patient health and give adequate care. Pathologists manually examine microscopic analyzation of blood smears by pathologists is inherently time-consuming, subjective, and susceptible to inter-observer inconsistencies. To overcome these constraints the study introduces a deep learning framework based on VGG19 transfer learning for binary of leukemia classification (Cancerous vs. Benign) from 3,256 microscopic blood cell images. A controlled experiment design compares Original (unprocessed) and Segmented (background-removed) image types under identical 3-fold stratified cross-validation conditions. The Original (Proposed) variant achieved superior sensitivity of 98.69%, accuracy of 97.39% , AUC-ROC of 0.9940 and F1 score of 0.9846. These results outperformed the Segmented variant on all cancer-detection-critical metrics. As per study 98.69% of sensitivity represents the highest reported, rigorously validated sensitivity scores in the comparable leukemia classification study. These study results demonstrate that contextual morphological information blood smear images provides valuable features effectively by VGG19's and effectively utilized by the ImageNet-pretrained by VGG19. Overall, the study demonstrated strong potential for clinical deployment as an initial screening system for detection of leukemia.

Keywords: Leukemia diagnosis, VGG19, transfer learning, deep learning, blood cell classification, image segmentation, k-fold cross-validation, sensitivity, medical imaging.

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Submission Files

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Submission Questions Response

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3. Author's Information

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I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

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7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

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